

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode in AI's reasoning with emem.

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From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source If one fails: refuse.	Reason with cited evidence Carry the reference onward.

1 Does agent B act on a corrupted handoff?

The same corruptions, handed over five ways.

Agent A cites a satellite reading. A relay changes the evidence. Agent B acts on it, declines it or uses the genuine record.

Agent A cites	handoff	relay	Agent B	B acted on corrupted evidence agents, pooled Claude	declined	used the genuine record
 Agent A reads NDVI 0.4709 and cites it Sentinel-2A L2A, 25 Sep 2026 outlined: the 10 m pixel the record names RGB from recorded grids	prose The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha...	relay or faulty signer changes: value cell time band source derivation stale state signature pixel	B reads the text	254 / 276	22	0
	JSON { "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448...		B reads the fields	164 / 276	106	6
	retrieved text (RAG) Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ...		B retrieves a passage	220 / 276	56	0
	opaque id record ref obs-7d1c44		B fetches the sender's record	154 / 288	88	46
	emem reference emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6...		B resolve re-hash place / time / band signature log recompute source re-read	0 / 300	264	36

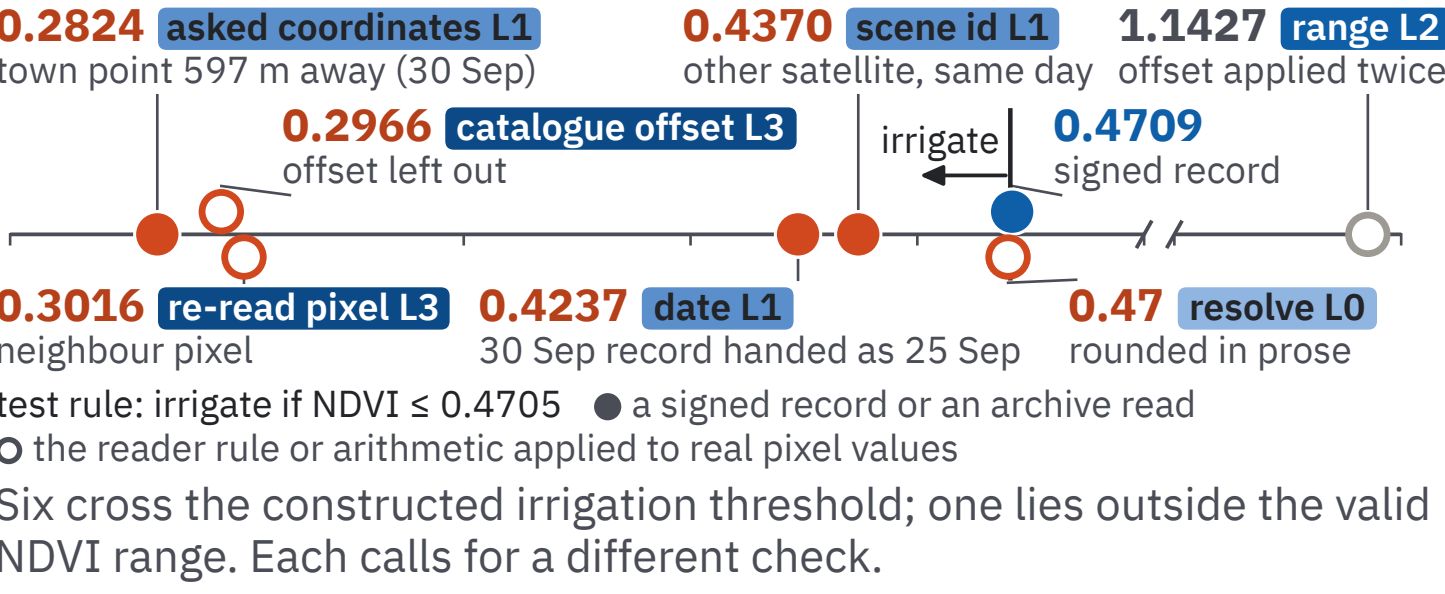
Across the 23 shared items, agents acted on 254 of 276 corrupted handoffs in prose, 154 of 276 with an opaque id, and 0 of 276 with an emem reference (2 of 276 without the instruction to check).

By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked into the AI's reasoning (SAT-042 harness, not flown).

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz : change in readout. The terms list possible causes (environment, sensor, geolocation, encoder, residual), not measured shares.

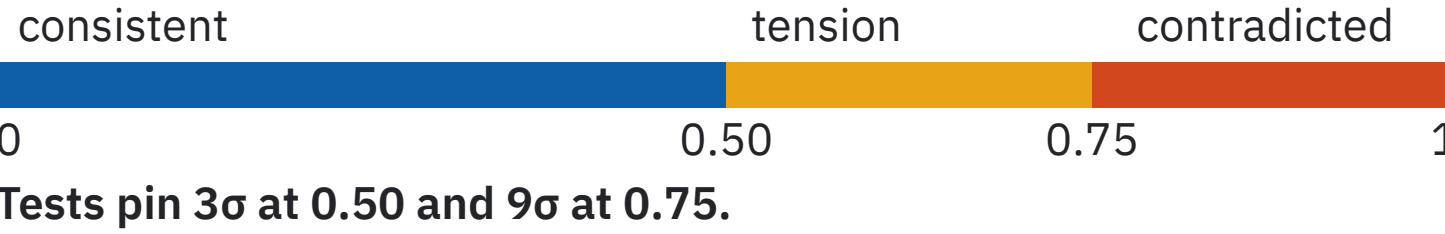
IMPLEMENTED HEURISTIC · DRIFT-ANCHOR SCORE

$$d = |x - a| \text{ for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma); s = r / (1 + r)$$

Invalid op: s = 0 if d = 0; otherwise s = 1 (fail-closed).

x: output · a: anchor · σ : anchor uncertainty



Tests pin 3σ at 0.50 and 9σ at 0.75.

Score boundaries: consistent < 0.50; tension < 0.75.

This score is an uncalibrated heuristic: it measures disagreement with one anchor, not a cause. SAT-042 uses it in the execution harness.

7 What does a checked reference establish?

The record, not physical truth

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token · **record** · **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(o) & leaf in STH & STHgeo.qa prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact, cid; one flipped mantissa bit gives mu5x2cks... · still trusted: BLAKE3 and the verifier's code

4-5 cell binding hashed cell = token cell = the lat/long the derivation used · the cell rule

6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code

8-9 upstream the STAC item matches the scene; the COG pixel at col 909, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum

10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right

12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI

15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI, DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 6).

12 Use emem. Carry the evidence forward.

Separately, the same record and value came back through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). Listings show availability; panel 1 measures agent decisions.

ChatGPT · @emem plugin “Signed facts about real places” Enable emem; paste a token.	Claude · Directory plugin / MCP “hand another agent a token it can check offline” Resolve from chat or terminal.	Dify · Marketplace plugin “Shared memory for AI agents. One address per fact, one signature you check.” Referent Lock / EUDR workflows	GitHub MCP Registry · Server listing “Satellites for AI.” Install MCP server	ClawHub · OpenClaw plugin listing “Shared, signed and content-addressed memory of the physical world...” Install the OpenClaw plugin.	CONNECT Setup · EO tasks Reproduce
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Measurements: 29 Sep to 1 Oct 2026. · Methods link each figure to its records, data and checks (Connect). · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)